

Java Basics - Object-Oriented Programming (OOP) Exam

Total Questions: 50 **Suggested Time:** 90 minutes **Instructions:**
Answer all questions. For multiple-choice questions, select the single best answer.

Part A - Multiple Choice (30 questions, 1 mark each)

- Which of the following best defines a **class** in Java?
 - A running instance of an object
 - A blueprint/template for creating objects**
 - A method that returns void
 - A built-in Java keyword
- What is an **object** in Java?
 - A syntax error
 - An instance of a class**
 - A type of loop
 - A reserved keyword
- Which keyword is used to create an object in Java?
 - class
 - new**
 - this
 - instance
- Which of these is NOT one of the four main pillars of OOP?
 - Encapsulation
 - Inheritance
 - Compilation**
 - Polymorphism
- What is **encapsulation**?
 - Wrapping data and methods together and restricting direct access to data**
 - Creating multiple classes with the same name
 - Converting one data type to another
 - Running two threads at once
- Which access modifier restricts a member to be accessible only within its own class?
 - public
 - protected
 - private**
 - default
- Which access modifier allows access from any other class in any package?
 - private
 - protected
 - public**
 - default** (no modifier)
- A constructor in Java:
 - Must always return a value
 - Has the same name as the class and no return type**
 - Can only be called once per program
 - Must be declared static

9. What is a **default constructor**?
 - a. A constructor with parameters
 - b. A no-argument constructor automatically provided by Java if no constructor is defined
 - c. A constructor that only exists in abstract classes
 - d. A constructor that must be private
10. Which keyword is used to refer to the current object inside a method or constructor?
 - a. self
 - b. this
 - c. current
 - d. obj
11. What does **inheritance** allow a class to do?
 - a. Store multiple data types in one variable
 - b. Acquire fields and methods from another class
 - c. Run without a main method
 - d. Avoid using constructors
12. Which keyword is used to inherit a class in Java?
 - a. implements
 - b. extends
 - c. inherits
 - d. super
13. In Java, a class can extend how many other classes directly?
 - a. Zero
 - b. One
 - c. Two
 - d. As many as needed
14. What is the purpose of the **super** keyword?
 - a. To create a new object
 - b. To refer to the parent class's members or constructor
 - c. To declare a class as final
 - d. To stop inheritance
15. What is **method overriding**?
 - a. Defining multiple methods with the same name but different parameters in the same class
 - b. Redefining a parent class method in a child class with the same signature
 - c. Declaring a method as private
 - d. Calling a method twice
16. What is **method overloading**?
 - a. Having multiple methods in the same class with the same name but different parameter lists
 - b. Overriding a method from a parent class
 - c. Declaring a method final
 - d. Using recursion
17. Which of the following can be overloaded in Java?
 - a. Only constructors
 - b. Only regular methods
 - c. Both methods and constructors
 - d. Neither
18. What is **polymorphism**?
 - a. The ability of an object to take many forms
 - b. The process of hiding data
 - c. A type of loop structure
 - d. A way to declare arrays
19. Which type of polymorphism is achieved through method overriding?
 - a. Compile-time (static) polymorphism
 - b. Run-time (dynamic) polymorphism
 - c. Constructor polymorphism

- d. None of the above
- 20. What is an **abstract class**?
 - a. A class that cannot have any methods
 - b. A class that cannot be instantiated directly and may contain abstract methods
 - c. A class with only static methods
 - d. A class that must implement an interface
- 21. Which keyword declares an abstract method?
 - a. final
 - b. static
 - c. abstract
 - d. private
- 22. What is an **interface** in Java?
 - a. A class with only constructors
 - b. A contract that specifies methods a class must implement
 - c. A type of loop
 - d. A private helper class
- 23. Which keyword does a class use to implement an interface?
 - a. extends
 - b. implements
 - c. inherits
 - d. uses
- 24. Can a Java interface (pre-Java 8 style) contain method bodies?
 - a. Yes, always
 - b. No, traditionally interface methods are abstract by default
 - c. Only in the main method
 - d. Only for private methods
- 25. What is the main difference between an abstract class and an interface?
 - a. There is no difference
 - b. An abstract class can have constructors and instance variables with state; an interface traditionally cannot
 - c. Interfaces can be instantiated directly
 - d. Abstract classes cannot have any methods
- 26. What does the **static** keyword mean when applied to a variable?
 - a. The variable belongs to the class, not to individual objects
 - b. The variable can never change
 - c. The variable is private by default
 - d. The variable must be initialized in a constructor
- 27. What does the **final** keyword do when applied to a variable?
 - a. Makes the variable static
 - b. Prevents the variable's value from being changed after initialization
 - c. Makes the variable public
 - d. Deletes the variable after use
- 28. What happens if a class is declared **final**?
 - a. It cannot be extended (no subclasses allowed)
 - b. It cannot have any methods
 - c. It must be abstract
 - d. It becomes an interface
- 29. Which of these correctly describes **constructor chaining** using `this()`?
 - a. Calling another constructor of the same class from within a constructor
 - b. Calling a method from another class
 - c. Declaring two constructors with the same signature
 - d. Making a constructor static
- 30. What is the output of calling a method on an object reference that is `null`?
 - a. It returns 0

- b. It throws a `NullPointerException`
 - c. It silently does nothing
 - d. It automatically creates a new object
-

Part B - True or False (10 questions, 1 mark each)

- 31. A static method in a subclass can override a static method in its parent class the same way instance methods are overridden. (True/False)
 - 32. A class can implement multiple interfaces at the same time. (True/False)
 - 33. Constructors can be overloaded just like regular methods. (True/False)
 - 34. Static methods can directly access non-static (instance) variables without an object reference. (True/False)
 - 35. An abstract class can have a constructor even though it cannot be instantiated directly. (True/False)
 - 36. Overriding requires the method name, return type (or covariant type), and parameter list to match the parent method. (True/False)
 - 37. In Java, all classes implicitly inherit from the `Object` class unless they already extend another class. (True/False)
 - 38. Encapsulation is typically achieved using private fields with public getter and setter methods. (True/False)
 - 39. Java supports multiple inheritance of classes (a class extending two classes at once). (True/False)
 - 40. A final method cannot be overridden by a subclass. (True/False)
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Part C - Short Answer / Code Output (10 questions, 2 marks each)

41. Write the output of the following code:

```
class Animal {
    void sound() {
        System.out.println("Animal makes a sound");
    }
}
class Dog extends Animal {
    void sound() {
        System.out.println("Dog barks");
    }
}
public class Main {
    public static void main(String[] args) {
        Animal a = new Dog();
        a.sound();
    }
}
```

```
    }
}
```

42. Explain in 2-3 sentences the difference between **overloading** and **overriding**.

same name different signature and even the return data type but overriding same name but difference in implementation

43. Write the output of the following code:

```
class Counter {
    static int count = 0;
    Counter() {
        count++;
    }
}

public class Main {
    public static void main(String[] args) {
        new Counter();
        new Counter();
        new Counter();
        System.out.println(Counter.count);
    }
}
```

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44. Given the class below, write ONE line of code to create an object of Car and call its drive() method:

```
class Car {
    void drive() {
        System.out.println("Car is driving");
    }
}
```

Car car = new Car();
car.drive();

45. What will happen if you try to instantiate the abstract class below directly with new Shape()? Explain why.

its error bcz the abstract class cant take an object from abstract class

```
abstract class Shape {
    abstract double area();
}
```

46. Write the output of the following code:

```
class Parent {
    Parent() {
        System.out.println("Parent constructor");
    }
}

class Child extends Parent {
    Child() {
        System.out.println("Child constructor");
    }
}

public class Main {
    public static void main(String[] args) {
        Child c = new Child();
    }
}
```

47. Identify and correct the error in the following code:

```
public class Main {
    private int x = 5;
    public static void main(String[] args) {
```

cant call x without getter bcz its private and its the defention of encapsoulation

```

        System.out.println(x);
    }
}

```

48. Briefly explain why encapsulation is considered good practice in OOP (2-3 sentences).

security and the privacy of data bcz you dont work

49. Write the output of the following alone and dont give an accees to anyone

```

interface Greeting {
    void sayHello();
}
class English implements Greeting {
    public void sayHello() {
        System.out.println("Hello!");
    }
}
public class Main {
    public static void main(String[] args) {
        Greeting g = new English();
        g.sayHello();
    }
}

```

the output

50. Write the output of the following code:

```

class Calculator {
    int add(int a, int b) {
        return a + b;
    }
    double add(double a, double b) {
        return a + b;
    }
}
public class Main {
    public static void main(String[] args) {
        Calculator c = new Calculator();
        System.out.println(c.add(2, 3));
        System.out.println(c.add(2.5, 3.5));
    }
}

```

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End of Exam